Zhaoxiang Feng

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Education

The University of Texas at Austin — Austin, TX
Master of Science in Computer Science — *Aug. 2026 – May 2028*

University of California San Diego — San Diego, CA
B.S. in Data Science; B.S. in Probability & Statistics; GPA: 3.84/4.00 — *Sep. 2022 – Mar. 2026*

Research Experience

Hao AI Lab, UC San Diego — San Diego, CA
Research Assistant, Advisor: Prof. Hao Zhang — *Sep. 2025 – Present*

- Built JetSpec's standalone **LLM inference engine**: **paged KV cache**, continuous batching, a **tree-attention Triton kernel**, and **CUDA-graph** verification, at ∼1,000 tok/s on one **NVIDIA B200**
- JetSpec reaches up to **9.64× end-to-end speedup** over autoregressive decoding on Qwen3-8B; second author, under review at NeurIPS 2026
- Led **JetSpec++**, an attention-layer causal correction for parallel drafting that accepts **0.89 more tokens per round** than DeepSeek's DSpark at matched size, widening to 1.67 at unseen draft depths
- Ported **DFlash** speculative decoding to **Google Cloud TPU** in **JAX** as primary contributor: **3.13×** average speedup over 9 benchmarks and **2.29× vs. 1.30×** for EAGLE-3; featured on the Google Developers Blog
- Upstreamed the port to **vLLM tpu-inference** in 3 PRs (model and proposer merged; integration in review)

Q-Lab, UC San Diego — San Diego, CA
Research Assistant, Advisor: Prof. Lianhui Qin — *Jan. 2025 – Jun. 2026*

- Contributed to **ArcMemo**, concept-level lifelong memory for LLM reasoning; **Runner-Up, ARC Prize 2025 Paper Awards** (top 8 of 90)
- Built a **concept-to-puzzle pipeline** in which an LLM turns each concept into executable generator code, filtered for determinism and well-formedness; adapted the method to **AIME** math

Wang Lab, UC San Diego — San Diego, CA
Student Researcher, Advisor: Prof. Wei Wang — *Jun. 2025 – Jun. 2026*

- Developing **TrustPPI**, a reliability signal for protein interaction models from embedding-perturbation stability: **0.66–0.83 AUROC** error detection across 3 architectures, vs. ∼0.5 for native confidence

Projects

LLM Pre-training System | *PyTorch, DDP, ZeRO-1, FlashAttention-2, torch.compile* — Sep. 2025 – Dec. 2025

- Built a training system for a **1.8B-parameter** model on **8× NVIDIA B200** as a UCSD senior capstone (Prof. Hao Zhang), with BF16 and FLOPs-accurate MFU monitoring
- Trained on **117B tokens** in a 60-hour budget at **160K tokens/s** and **36.7% MFU**; added SFT and DPO post-training (up to +2.6% on benchmarks)

Technical Skills

Languages: Python, Triton, R, SQL, Java, JavaScript
ML Systems: vLLM, speculative decoding, paged KV cache, continuous batching, CUDA graphs, FlashAttention-2
Hardware & Distributed: NVIDIA B200/H100, Google Cloud TPU (v5p, v6e), JAX, PyTorch DDP, ZeRO
ML & Tools: PyTorch, Hugging Face Transformers, SFT, DPO, RAG, Git, Linux, Docker